Project 3: Ion Drive Rocket Analysis

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The purpose of this project is to analyze an ion drive propelled, free space rocket, with a Brayton cycle power plant. Given the total mass (80,000 kg) and payload mass (5,000 kg), the required propellant and engine mass are found. The engine mass and other given values are used to calculate the ion drive performance and power requirements. The power required by the ion drive is then used to in designing the required Brayton cycle power plant. By knowing the engine performance and onboard propellant the time of flight and distance traveled can be found numerically.

Nomenclature

T	=	temperature $[K]$
W	=	power $[Watt]$
Q	=	heat addition $[Watt]$
A	=	area $\left[m^2\right]$
F	=	thrust $[N]$
R	=	gas constant $[J/kgK]$
C_p	=	constant pressure specific heat $[J/kgK]$
γ	=	specific heat ratio
$\dot{m}$	=	mass flow rate $[kg/s]$
I_{sp}	=	specific impulse $[s]$
g_0	=	standard gravitational acceleration $\left[9.81m/s^2\right]$

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I. Rocket Analysis

The following equation can be used to calculate the propellant mass required to perform a maneuver.

$$\frac{\Delta v}{C} = \ln\left(\frac{m_0}{m_0 - m_p}\right) - \int_{t_0}^{t_f} \frac{D}{Cm} dt - \int_{t_0}^{t_f} \frac{g\cos\theta}{C} dt \tag{1}$$

Because the drag term and the gravity can be ignored, this equation can be simplified to the following.

$$m_p = m_0\left(1 - e^{\frac{-\Delta v}{C}}\right) \tag{2}$$

C in this equation is the effective exhaust velocity which can be found by using the following.

$$C = I_{sp} g_0 = 34335.0\ m/s \tag{3}$$

The required propellant mass for each stage of the flight is:

$$m_{p_a} = \boxed{51,134.167\ kg}$$

$$m_{p_d} = \boxed{18,450.379\ kg}$$

This gives a total propellant mass of:

$$m_p = 69584.546\ kg$$

Using the assumption that after the deceleration all the propellant is spent (the only remaining mass is that of the engine and payload), knowledge of the initial mass and payload mass can be used to calculate the engine mass. The engine mass m_e can be calculated using the equation below. Here m_0 is the initial mass, m_p is the propellant mass, and m_L is the payload mass.

$$m_e = m_0 - (m_p + m_L) = \boxed{5415.454\ kg} \tag{4}$$

This engine mass is then used in subsequent ion drive calculations.

II. Ion Drive Analysis

Using the engine mass calculated previously the total engine power, accelerator and ionizer, can be calculated given that the engine power per unit engine mass is 600 Watts per kilogram.

$$W_{engine} = \boxed{3,249,272\ W}$$

Since it is known that the ionizer uses 35% of the power that the accelerator uses, the accelerator power can be found by using the following:

$$W_{accelerator} = W_{ion} = \frac{W_{engine}}{1 + 35/100} = 2,406,868\ W \tag{5}$$

This gives an accelerator efficiency of:

$$\eta_{accelerator} = \frac{W_a}{W_{engine}} \times 100\% = \boxed{74.074\%} \tag{6}$$

The effective exhaust velocity in the specific impulse equation can be rewritten in terms of propellant charge per mass and potential difference. Solving for the potential difference gives the voltage required by the accelerator.

$$I_{sp} = \frac{1}{g_0}\sqrt{2\frac{q}{m}(\phi_A - \phi_C)} \rightarrow (\phi_A - \phi_C) = V = \boxed{235,778\ V} \tag{7}$$

Using the accelerator power and voltage, the required current can be calculated:

$$I = \frac{W_{ion}}{V} = \boxed{10.208\ A} \tag{8}$$

Using the maximum allowable voltage gradient the distance between the anode and cathode can be found:

$$\frac{(\phi_A - \phi_C)}{d} = 10^5\ \frac{V}{cm} \rightarrow d = \boxed{2.35778\ cm} \tag{9}$$

To find the beam area the equation for current per cross sectional area can be used. Because the assumption that the maximum possible current is being used the following can be done:

$$j_{max} = 0.4444\epsilon_0\sqrt{2\frac{q}{m}}\frac{(\phi_A - \phi_C)^{3/2}}{d^2} = 57.273\ \frac{A}{m^2} \tag{10}$$

Where ϵ_0 is the universal constant of permittivity of free space $8.85 \times 10^{-12}\ \frac{C^2}{Nm^2}$

The area is then:

$$A_{beam} = \frac{I}{j_{max}} = 1782.37\ cm^2 = 0.178237\ m^2 \tag{11}$$

The beam diameter is:

$$D_{beam} = \sqrt{\frac{\pi A_{beam}}{4}} = \boxed{0.47638\ m}$$

Following this, and using the same maximum current assumption, the thrust can be found:

$$T = F = 0.8888\epsilon_0 \left(\frac{\phi_A - \phi_C}{d}\right)^2 A_{beam} = \boxed{140.20\ N} \tag{12}$$

Now that thrust is known and the effective exhaust velocity was found previously, the mass flow through the engine can be found:

$$T = \dot{m}C \rightarrow \dot{m} = \boxed{0.00408327\frac{kg}{s} = 4.08327\ \frac{g}{s}} \tag{13}$$

III. Brayton Cycle Analysis

First the gas constant and specific heat of diatomic nitrogen was found. R_u is the universal gas constant and M is the molecular weight.

$$R = \frac{R_u}{M} = 296.945\ \frac{J}{kgK} \tag{14}$$

$$C_p = R\frac{\gamma}{\gamma - 1} = 1039.307\ \frac{J}{kgK} \tag{15}$$

Following this a mesh of all possible compressor ratios and compressor inlet temperatures was made. This mesh was iterated through until the minimum physical area was found. Solutions resulting in a negative area were removed. The following work shows the values for the minimum radiator area solution.

First, guess a compressor pressure ratio and radiator exit temperature:

$$\pi_c = \boxed{2.635}$$

$$T_a = \boxed{1473.347\ K}$$

The turbine pressure ratio is the inverse of the compressor pressure ratio:

$$\pi_t = \frac{1}{\pi_c} = 0.37946 \tag{16}$$

Temperature at the radiator entrance/turbine exit:

$$\frac{T_4}{T_3} = 1 - \eta_t \left(1 - \pi_t^{\frac{\gamma-1}{\gamma}}\right) \rightarrow T_4 = \boxed{2347.048\ K} \tag{17}$$

Temperature at the compressor exit:

$$\frac{T_2}{T_a} = 1 + \frac{1}{\eta_c}\left(\pi_c^{\frac{\gamma-1}{\gamma}} - 1\right) \rightarrow T_2 = \boxed{1995.521\ K} \tag{18}$$

Thermal efficiency:

$$\eta = 1 - \frac{T_4 - T_a}{T_3 - T_2} = \boxed{0.1301944} \tag{19}$$

Using the power required for the ion drive and the calculated efficiency and temperatures, the area of the radiator can be calculate. σ is the Boltzmann constant of radiation ($5.67 \times 10^{-8}\ \frac{W}{m^2K^4}$), and ϵ is the emissivity of the radiator which is assumed to be 1.

$$\frac{A_{rad}}{W_{out}} = \frac{A_{rad}}{W_{engine}} = \frac{\left(\frac{1}{\eta} - 1\right)}{3(T_4 - T_a)\sigma\epsilon}\left(\frac{1}{T_a^3} - \frac{1}{T_4^3}\right) \rightarrow A_{rad} = \boxed{34.3705\ m^2} \tag{20}$$

The required heat addition required by the reactor is just the work required divided by the thermal efficiency:

$$Q = \frac{W}{\eta} = \boxed{24,957,085\ W} \tag{21}$$

The required mass flow through the cycle is:

$$\dot{m}_{gen} = \frac{\frac{1}{\eta} - 1}{T_4 - T_a}\frac{W_{out}}{C_p} = \boxed{23.906\ \frac{kg}{s}} \tag{22}$$

The power required for the compressor is:

$$W_c = \dot{m}_{gen} C_p\ (T_2 - T_a) = \boxed{12,973,853\ W} \tag{23}$$

The power provided by the turbine is:

$$W_t = \dot{m}_{gen} C_p\ (T_3 - T_4) = \boxed{16,223,126\ W} \tag{24}$$

The difference of the two returns the power required to run the ion drive.

IV. Rocket Trajectory

The time of flight for the acceleration and deceleration phase of the flight can be calculated using the engine mass flow rate and the propellant required for that phase:

$$TOF = \frac{m_p}{\dot{m}_{ion}} \tag{25}$$

The time required for acceleration is:

$$TOF_{acc} = \boxed{12,522,843\ seconds = 144.940\ days}$$

The time required for deceleration is:

$$TOF_{dec} = \boxed{4,518,529\ seconds = 52.298\ days}$$

The procedure for integrating the velocity goes as shown:

Create lists with the mass and time from beginning to end of the phase of flight

Set the initial velocity for the phase V_0 to V_i and set distance equal to zero

Begin loop:

calculate the change in mass from the current time step to the next:

$$dm = m_{i+1} - m_i \tag{26}$$

Use the change in mass to calculate the change in velocity:

$$dV = -C\frac{dm}{m_i} \tag{27}$$

Add the change in velocity to the current velocity:

$$V_{i+1} = V_i + dV \tag{28}$$

Calculate the change in time:

$$dt = t_{i+1} - t_i \tag{29}$$

Add the differential distance:

$$Distance_{i+1} = Distance_i + V_{i+1}dt \tag{30}$$

Repeat for all time steps.

During the acceleration phase the spacecraft travels:

$$Distance = \boxed{1.82547 \times 10^{11}\ m}$$

During cruise the spacecraft travels:

$$Distance = \boxed{1.20960 \times 10^{13}\ m}$$

During the deceleration phase the spacecraft travels:

$$Distance = \boxed{9.22815 \times 10^{10}\ m}$$

The total distance traveled is:

$$Distance = \boxed{1.23708 \times 10^{13}\ m \approx 82.7 AU}$$

V. Results

Ion Drive		
Current	A	10.308
Voltage	kV	235.8
Acceleration Distance	cm	2.357
Prop Mass Flow	g/s	4.083
Thrust	N	140.2
Beam Diameter	m	0.476
Required Power	MW	3.249
Accelerator Efficiency	%	74.074

Brayton Cycle		
Compressor Pressure Ratio		2.635
Compressor Inlet Temperature	K	1473.347
Compressor Exit Temperature	K	1995.521
Turbine Exit Temperature	K	2347.048
Turbine Power Supplied	MW	16.223
Compressor Power Required	MW	12.973
Reactor Heat Rate	MW	24.957
Thermal Efficiency	%	13.019
Radiator Area	m^2	34.371
Nitrogen Mass Flow Rate	kg/s	23.906

Free Space Rocket Analysis		
Engine Mass	kg	5415.45
Acceleration Prop Mass	kg	51134.2
Deceleration Prop Mass	kg	18450.4
Acceleration Time	days	144.94
Deceleration Time	days	52.298
Acceleration Distance	m	1.83E+11
Cruise Distance	m	1.21E+13
Deceleration Distance	m	9.23E+10
Total Distance	m	1.24E+13
Total Distance	AU	82.7

Appendix A

```
import numpy as np
import matplotlib.pyplot as plt
import ion_drive_solver as ion_drive

Isp = 3500 # sec
prop_charge = 2500 # C/kg
max_V_gradient = 1.0e+5
drive_power = 0.35 # percent/100

m_0 = 80_000 # Kg
m_L = 5000 # Kg

alpha_prime = 600 # W/kg

# # Trajectory

# %%
g0 = 9.81
C = Isp * g0 # effective velocity

dV_accel = 35_000
dV_deccel = 35_000
dV_tot = dV_accel + dV_deccel



m_p_func = lambda dV, m_0: m_0 * (1-np.exp(-dV/C))

m_p_accel = m_p_func(dV_accel,m_0)
m_p_deccel = m_p_func(dV_deccel,m_0-m_p_accel)
m_p_tot = m_p_accel + m_p_deccel

engine_mass = m_0 - m_L - m_p_tot

# # Ion Drive
epsilon0 = 8.85e-12

W_engine = engine_mass * alpha_prime

W_ion =  W_engine / (1 + drive_power)
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```
W_ionizer= W_ion * drive_power

voltage = (Isp * g0)**2 / (2 * prop_charge)
current = W_ion / voltage
acc_distance_cm = voltage / max_V_gradient
d = acc_distance_cm/100

j_max = 0.4444 * epsilon0 * np.sqrt(2*prop_charge) * voltage**(3/2) / d**2

A_beam = current/j_max
D_beam = np.sqrt(A_beam * 4 / np.pi)
F = 0.8888 * epsilon0 * (voltage/d)**2 * A_beam
m_dot_ion = F/C

# # Brayton cycle
Ru = 8314.4598 # J/mol/K
# nitrogen
molar_mass = 28 # kg/kmol
gamma = 1.4
R = Ru/molar_mass
Cp = gamma*R/(gamma-1)

T3 = 3000 # K
Nsize = 500
pi_c_list = np.linspace(1,9,Nsize)
Ta_list = np.linspace(800,1600, Nsize)

pi_c_list, Ta_list = np.meshgrid(pi_c_list,Ta_list)
pi_c_list = pi_c_list.flatten()
Ta_list = Ta_list.flatten()
eta_c = 0.90
eta_t = 0.90

epsilon = 1.0
sigma = 5.670374419e-8

def brayton_cycle(pi_c, Ta):
    pi_t = 1/pi_c
    T4 = T3 * (1 - eta_t * (1 - pi_t**((gamma-1)/gamma)))
    T2 = Ta * (1 + 1/eta_c * (pi_c**((gamma-1)/gamma) - 1))
    eta = 1 - (T4 - Ta)/(T3 - T2)
```

```
    Arad = (1/eta - 1)/(3*(T4 - Ta)*sigma*epsilon) * (1/Ta**3 - 1/T4**3)
    return T2,T4,eta,Arad

T2,T4,eta,Arad = brayton_cycle(pi_c_list,Ta_list)
for i in range(pi_c_list.size):
    if Arad[i] <= 0:
        Arad[i] = np.nan
Arad2 = Arad[~np.isnan(Arad)]
idx = np.where(Arad == np.min(Arad2))

Ta = Ta_list[idx][0]
pi_c = pi_c_list[idx][0]
T2 = T2[idx][0]
T4 = T4[idx][0]
eta = eta[idx][0]
Arad = Arad[idx][0]



min_rad_area = Arad*W_engine
Q_added = W_engine/eta
m_dot = (1/eta-1)/(T4-Ta)/Cp* W_engine

W_c = m_dot * Cp * (T2-Ta)
W_t = m_dot * Cp * (T3-T4)
net_power = W_t - W_c

# # Rocket Trajectory
TOF1 = m_p_accel / m_dot_ion#/3600/24
TOF2 = m_p_deccel / m_dot_ion#/3600/24

g0 = 9.81
C = Isp * g0 # effective velocity

dV_accel = 35_000
dV_deccel = 35_000
dV_tot = dV_accel + dV_deccel

N_itr = 100_000_00

m_accel_list = np.linspace(m_0,m_0-m_p_accel,N_itr+1)
t_accel_list = np.linspace(0,TOF1,N_itr+1)
```

```
def distance(V0,TOF,m0,mp,N_itr = 10_000_000,accel = True):
    m_list = np.linspace(m0,m0-mp,N_itr+1)
    t_list = np.linspace(0,TOF,N_itr+1)

    mult = 1
    if accel == False:
        mult = -1
    Vi = V0
    Dist = 0
    for i in range(N_itr):
        dt = t_list[i+1] - t_list[i]

        mi = m_list[i]
        dm = m_list[i+1] - mi
        dv = -C * (dm/mi) * mult
        Vi += dv
        Dist += Vi * dt
    print(Dist)


print(f'{TOF1 = }')
print(f'{TOF2 = }')

distance(0,TOF1,m_0,m_p_accel,N_itr=20_000_000)
TOF_cruise = 4000*24*3600
distance(35_000,TOF_cruise,m_0-m_p_accel,0,N_itr=20_000_000)
distance(35_000,TOF2,m_0-m_p_accel,m_p_deccel,N_itr=20_000_000,accel=False)
```